

Qinnan Jiang

To24157@bristol.ac.uk | [+44 7419 331076] | Bristol, United Kingdom | github.com/nanj-robotics | nanj-robotics.github.io

PROFILE

Electrical and Electronic Engineering undergraduate at the University of Bristol with an average of **83% (First-Class)** in second year. Hands-on robotics engineer working across mobile navigation, robot manipulation, visual perception, and reinforcement learning. Seeking graduate research in robotics, robot perception, robot learning and embodied intelligence.

EDUCATION

University of Bristol — Bristol, United Kingdom

2024 – Expected 2027

BEng (Hons) Electrical and Electronic Engineering | On track for First Class (UK First threshold = 70)

Year 2 (2025/26) (Average 83%)

Module	Credits	Mark
Electromagnetics and Semiconductor Devices	20	99
Signals and Communications	20	95
Engineering Mathematics 2	20	82
Electrical and Electronic Engineering Practice	40	76
Electrical Energy Conversion and Supply	20	70

ROBOTICS RESEARCH & INDUSTRY EXPERIENCE

PPO for Locomotion Shanghai Immersion Information Technology Co., Ltd. | Aug 2025 - Oct 2025 | nanj-robotics.github.io

SolidWorks · Isaac Gym · Reinforcement Learning · PPO · Robot Learning · Locomotion

- Modelled a 20-DoF custom wheeled-legged quadruped assembly in SolidWorks and exported the URDF.
- Trained a PPO locomotion policy in Isaac Gym (4,096 parallel GPU envs), achieving omni-directional locomotion and velocity tracking on flat and rough terrain.
- Designed a dense multi-term reward function with domain randomization (motor latency, friction, mass, push disturbances, observation noise) and a flat → rough → stair curriculum for sim-to-real transfer.
- Achieved 1.0 m/s forward speed, >95% velocity-tracking success; converged after 5,000 iterations (~2.0 hours on RTX 4090).

Dual 7-DoF Robot Arm Beijing SunstepVision Co., Ltd. | Jul 2026 - Sep 2026 | github.com/nanj-robotics/RoboMantis

C++ / Python · ROS 2 Humble · MoveIt2 · ros2_control · YOLOv11-Seg · FoundationPose · CAN bus · Pinocchio

- Designed and built a custom dual 7-DoF manipulator end to end: SolidWorks-to-URDF model, dual hardware interfaces, MoveIt2 dual-arm planning configuration and the grasp-execution node.
- Built the full perception-to-planning chain: YOLOv11-Seg instance segmentation → FoundationPose 6D pose estimation → eye-to-hand calibration → camera-to-base transform → MoveIt2 planning, reaching 90% grasp success over 20 trials.
- Collected and labelled a 100-image dataset with RGB-D camera, and trained YOLOv11-Seg with Ultralytics/PyTorch.
- Implemented gravity-compensated zero-torque (kinesthetic-teaching) controllers on both arms with Pinocchio RNEA dynamics and adaptive damping, enabling manual dragging for demonstration and VLA data collection.

Dual-Motor Synchronous Control Beijing SunstepVision Co., Ltd. | Aug 2026 | github.com/nanj-robotics/dual_motor_sync

C++ · ROS 2 Humble · CAN bus · trajectory planning · real-time control

- Authored a ROS 2 C++ package synchronizing two mirrored motors over CAN using time-parameterised cosine trajectories and 500 Hz back-to-back frame transmission in MIT PD-torque mode.
- Achieved synchronisation error down to 0.03° RMS.

Autonomous Mobile Robot Beijing SunstepVision Co., Ltd. | Jun 2026 - Jul 2026 | github.com/nanj-robotics/geni_rover
Python / C++ · ROS 2 Humble · Nav2 · SLAM/AMCL · EKF · RTK-GNSS · NvBlox · NVIDIA Jetson AGX Orin

- Developed a four-wheel differential-drive robot supporting both indoor navigation and outdoor RTK-GNSS navigation, deployed on an NVIDIA Jetson AGX Orin.
- Designed a dual-EKF localization architecture fusing wheel odometry, a 9-axis IMU, 2D LiDAR and a dual-antenna RTK GNSS receiver.
- Implemented the GNSS driver and NTRIP RTCM correction bridge for centimetre-level global positioning.
- Built 3D obstacle avoidance from RGB-D camera using nvblox ESDF costmap layers, and tuned the Nav2 stack with MPPI controller for real-world navigation.

ACADEMIC & UNIVERSITY PROJECTS

FPGA Peak Detection Prototyping University of Bristol | Jan 2026 – Apr 2026
VHDL · RTL Design · Digital System Design · FPGA

- Team-based hardware design project, implemented a peak-detection system on FPGA using VHDL RTL design.
- Developed an FSM controller and two-way handshaking protocol to synchronize high-speed FPGA logic with UART communication, ensuring zero data loss.
- Team project grade: 94.1%

Fan-Duct & Heatsink Thermal Characterisation University of Bristol | Jan 2025 - Apr 2025
Fusion 360 · 3D Printing

- Worked within a team on an electromechanical cooling-system project, covering mechanical design, numerical modelling, prototyping and experimental validation.
- Completed 3D modelling of fan-duct hardware using Fusion 360 and model prototype fabrication via 3D printing.
- Developed parametric thermal modelling workflows using MATLAB and Python, implementing iterative computation to predict heatsink thermal-resistance performance.
- Participated in labs to characterise heatsink behaviour and measure LED efficiency; analysed experimental data and contributed to the final group technical report.

TECHNICAL SKILLS

Programming	Python, C++, C, MATLAB, VHDL (digital/FPGA design)
Robotics	ROS/ROS2, MoveIt2, ros2_control, Nav2, SLAM, AMCL, Pinocchio, RViz, URDF/Xacro, Gazebo
Perception & ML	YOLO, FoundationPose, OpenCV, nvblox, PyTorch, CUDA, LabelMe
Control & Estimation	PID, MIT-mode PD/torque control, cosine/S-curve trajectory generation, Pinocchio RNEA gravity compensation, EKF fusion
Embedded & Hardware	CAN bus / SocketCAN, LiDAR, RTK GNSS, 9-axis IMU, RGB-D camera, NVIDIA Jetson AGX Orin
Tools	Linux/Ubuntu, Git, conda (Miniconda), SolidWorks, Fusion 360, CMake, colcon, LabelMe

ADDITIONAL

Languages	Mandarin Chinese (native); English (fluent — entire degree taught and examined in English)
Interests	Embodied intelligence, robot manipulation, legged robot, mobile robot